Experiment No 10

Experiment Title: Importing different types of datasets in Tableau and Dashboard Creation

Objective:

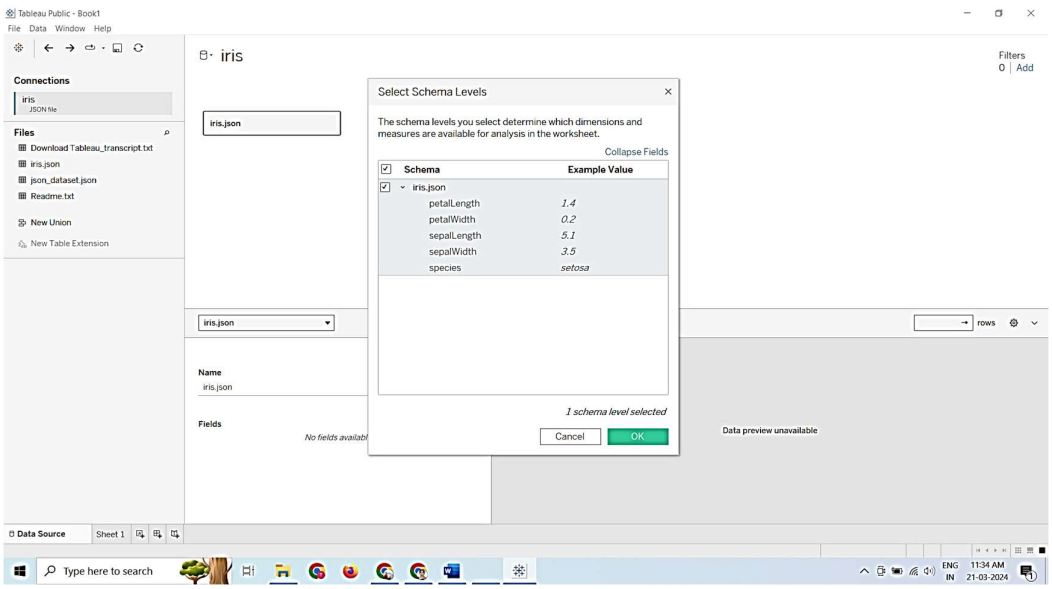
1. Import different types of datasets in Tableau
2. Design and create a basic dashboard in Tableau that presents key insights from the dataset.

Theory:

Loading Dataset in JSON/CSV in Tableau Public

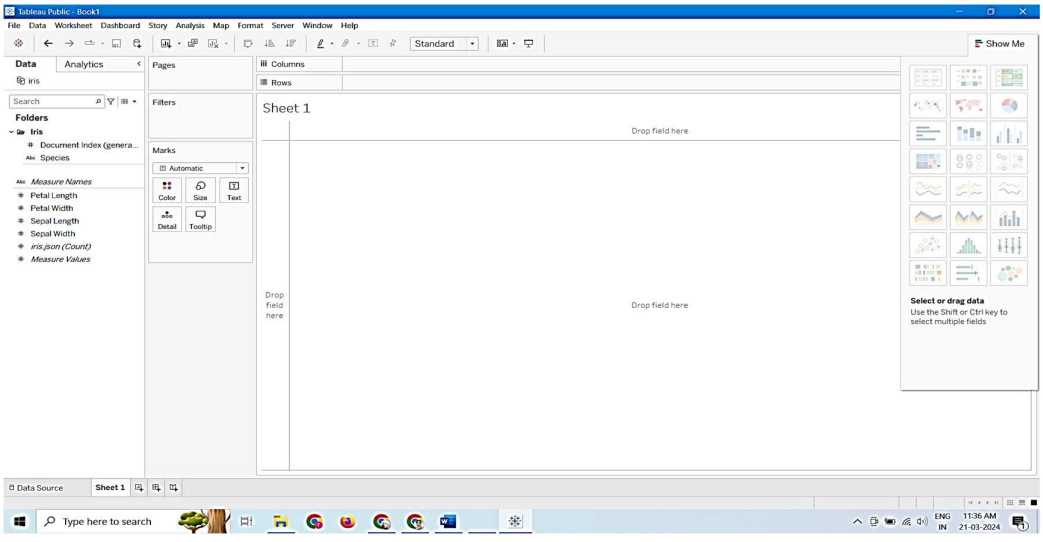
Click on JSON in Connect Tab

Select the JSON Dataset file. Window select Schema levels opens as given below.



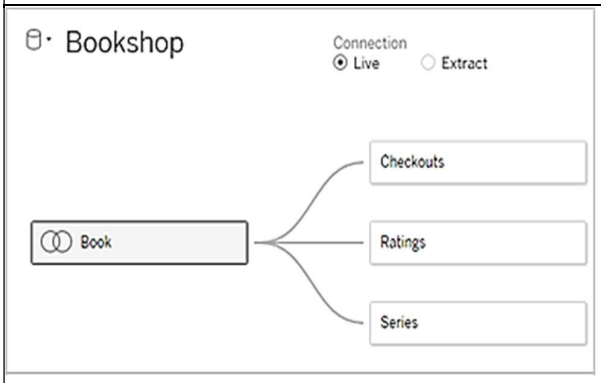
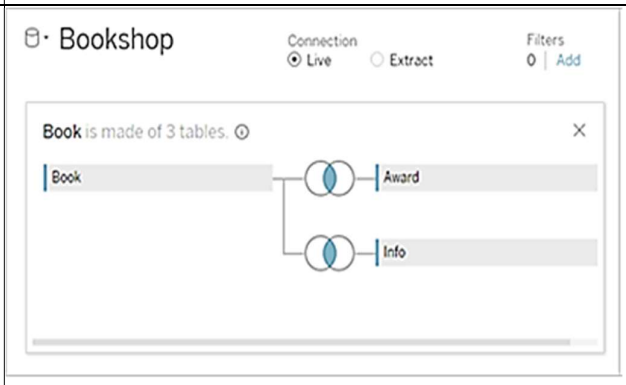
Click on OK

It will load the tables and shows the data types and contents of table.



The data model has two layers:

- The default view that you first see in the Data Source page canvas is the *logical layer* of the data source. You combine data in the logical layer using relationships (or noodles).
- The next layer is the *physical layer*. You combine data between tables at the physical layer using joins(Link opens in a new window) and unions. Each logical table contains at least one physical table in this layer. Think of the physical layer as the Join/Union canvas in the Data Source page. Double-click a logical table to view or add joins and unions.

LOGICAL LAYER	PHYSICAL LAYER
Noodles = Relationships	Venn diagram = Joins
	
The top-level view of a data source with multiple, related tables. This is the logical layer. Logical tables can be combined using relationships (noodles). They don't use join types. They act like containers for physical tables.	Double-click a logical table to open it and see its physical tables. Physical tables can be combined using joins or unions. In this example, the Book logical table is made of three, joined physical tables (Book, Award, Info).
Relationships canvas in the Data Source page	Join/Union canvas in the Data Source page
Tables that you drag here are called logical tables	Tables that you drag here are called physical tables
Logical tables can be related to other logical tables	Physical tables can be joined or unioned to other physical tables
Logical tables are like containers for physical tables	Double-click a logical table to see its physical tables
Level of detail is at the row level of the logical table	Level of detail is at the row level of merged physical tables

Layers of the data model

The top-level view that you see of a data source is the logical layer of the data model. You can also think of it as the Relationships canvas, because you combine tables here using relationships instead of joins.

When you combine data from multiple tables, each table that you drag to the canvas in the logical layer must have a relationship to another table. You do not need to specify join types for relationships; during analysis Tableau automatically selects the appropriate join types based on the fields and context of analysis in the worksheet.

The physical layer of the data model is where you can combine data using joins and unions. You can only use pivots in this canvas. You can think of it as the Join/Union canvas.

Build a new model

When you add one or more tables to the logical layer, you are essentially building the data model for your data source. A data source can be made of a single, logical table, or you can drag multiple tables to the canvas to create a more complex model.

- The first table that you drag to the canvas becomes the root table for the data model in your data source.
- After you drag out the root table, you can drag out additional tables in any order. You will need to consider which tables should be related to each other, and the matching field pairs that you define for each relationship.
- If you are creating a star schema, it can be helpful to drag the fact table out first, and then relate dimension tables to that table.
- Deleting a table in the canvas automatically deletes its related descendants as well. If you delete the root table, all other tables in the model are also removed.
- Each relationship must be made of at least one matched pair of fields. Add multiple field pairs to create a compound relationship. Matched pairs must have the same data type. Changing the data type in the Data Source page does not change this requirement. Tableau will still use the data type in the underlying database for queries.
- Relationships can be based on calculated fields.
- You can specify how fields used in the relationships should be compared by using operators when you define the relationship

Supported data model schemas

The data modeling capabilities introduced to Tableau in 2020.2 are designed to make analysis over common multi-table data scenarios—including star and snowflake data models—easy. The following types of models are supported in Tableau data sources.

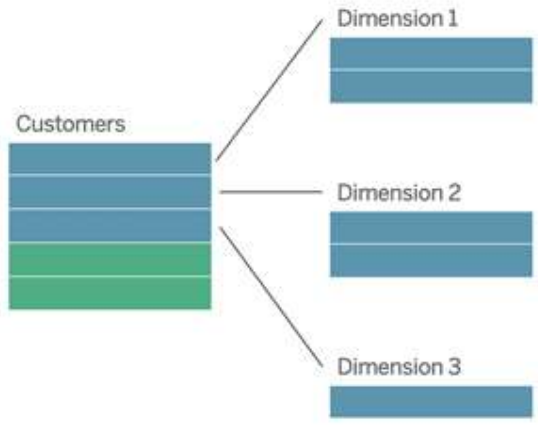
Single-table

Analysis over a single logical table that contains a mixture of dimensions and measures works just as in Tableau pre-2020.2. You can build a logical table using a combination of joins, unions, custom SQL, and so on.



Star and snowflake

In enterprise data warehouses, it is common to have data structured in star or snowflake schemas where measures are contained in a central fact table and dimensions are stored separately in independent dimension tables. This organization of data supports many common analysis flows including rollup and drill down.



These models can be directly represented with relationships in the data modeling capabilities available starting with Tableau 2020.2.

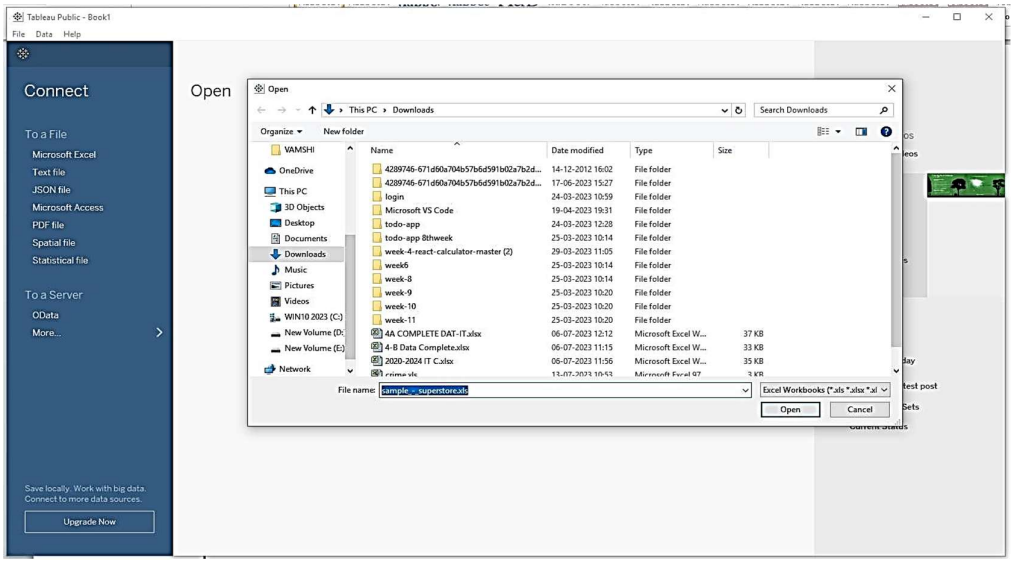
Drag the fact table into the model first and then relate the dimension tables to the fact table (in a star schema) or to other dimension tables (in a snowflake).

Typically, in a well-modeled star or snowflake schema, the relationships between the fact table and the dimension tables will be many-to-one.

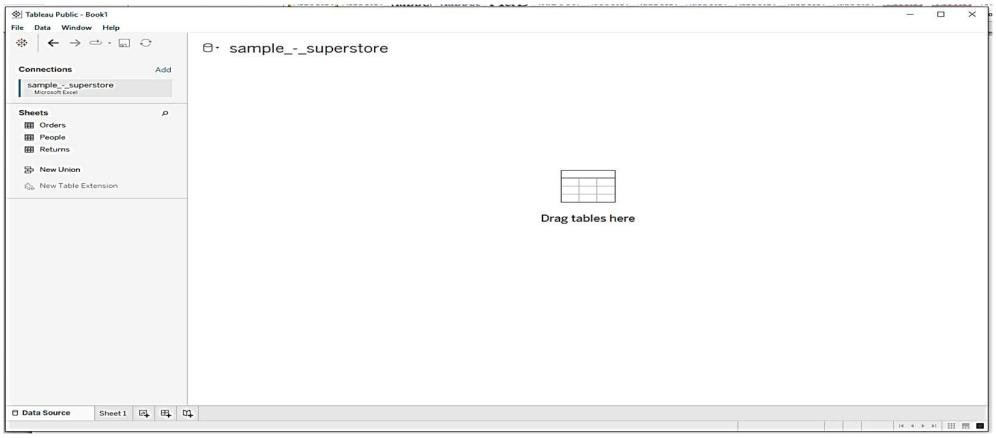
In a well-modeled star or snowflake schema, every row in the fact table will have a matching entry in each of the dimension tables. If this is true and captured in your data warehouse integrity constraints, Tableau will automatically use this information to set the referential integrity setting in Performance Options. If some fact table rows do not have a matching row in a dimension table (sometimes called “late-arriving dimensions” or “early-arriving facts”), Tableau will default to retaining all rows when computing measures, but may drop values when showing dimension headers.

Connecting to Data and preparing data for visualization in Tableau

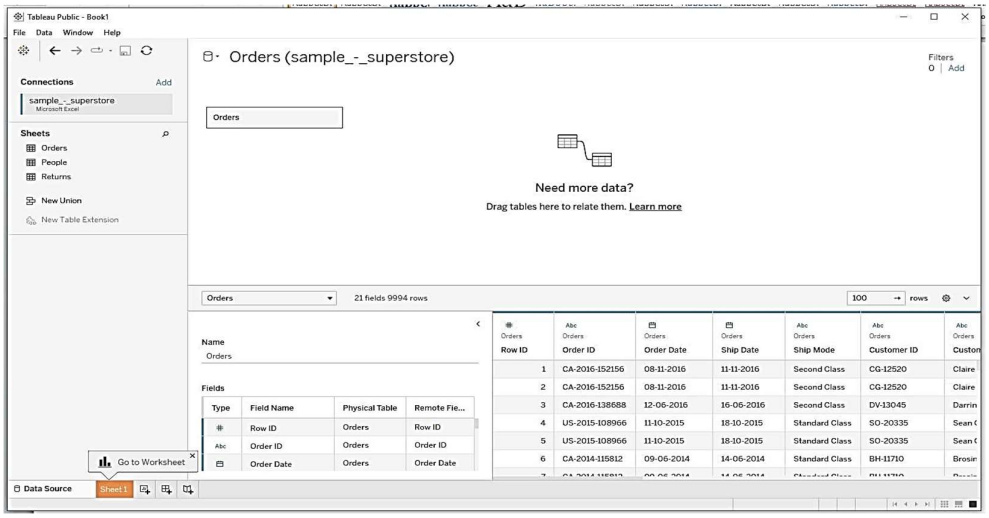
Data can be imported in Tableau Public from Connect panel on left side. For example, an Excel sample data set was loaded into Tableau as follows:



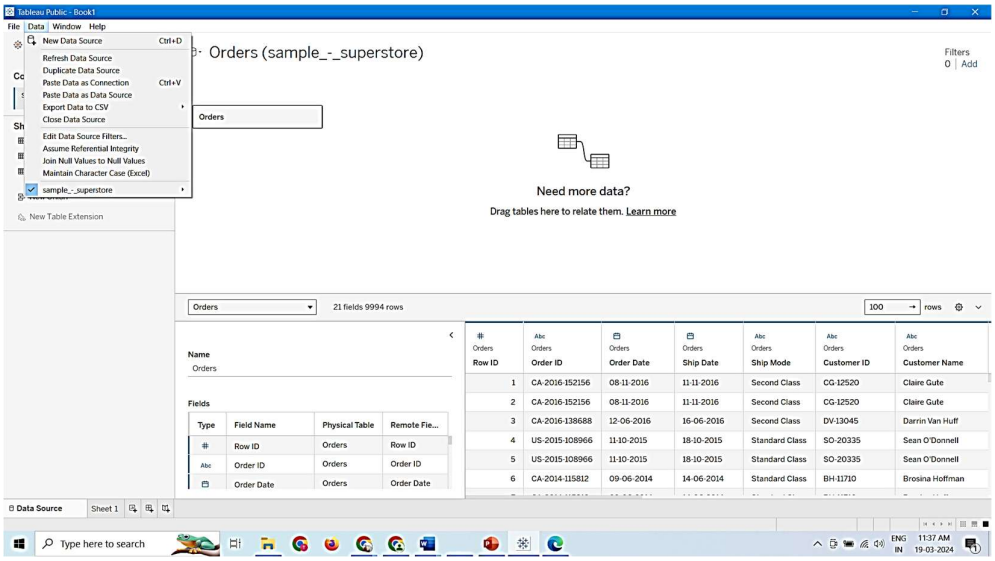
After clicking on open, screen is as follows: The data store page appears as above. The left pan shows that above dataset consists of 3 worksheets. If we drag orders table, screen appears as



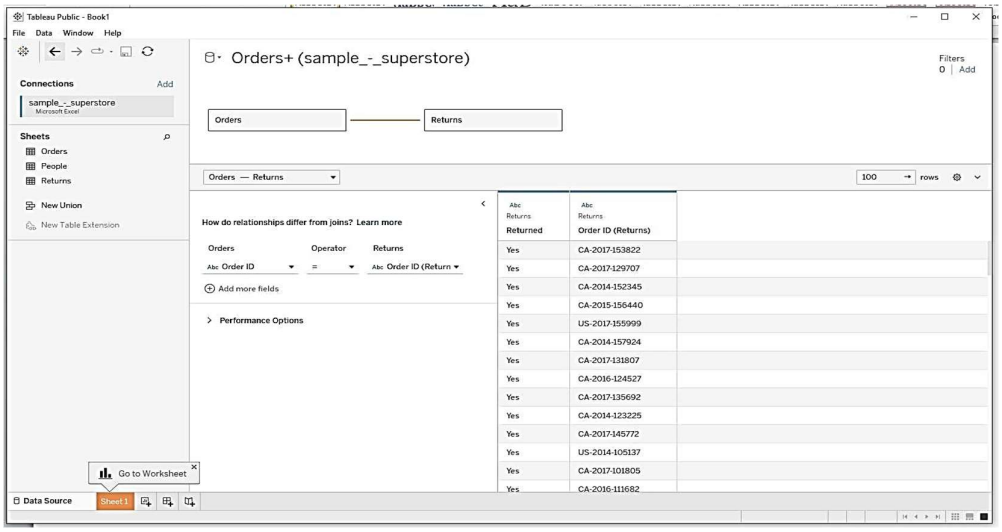
follows: Tableau automatically identifies the data type of each column.



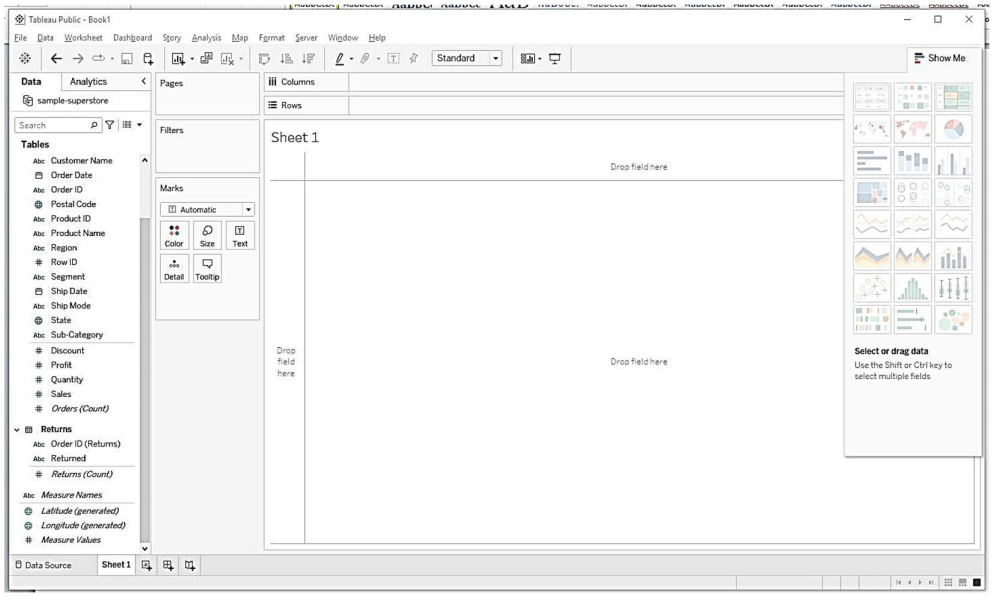
Now drag Returns table onto the Canvas to the right of Orders table. This shows the relation between the two tables Orders and Returns.



We can edit data source filters. We can duplicate Data Source. Join null Values to Null Values option is available under Data Menu.



If we click on the link between Orders and Returns table names at the top gives the summary of the relationship between the tables. Now rename the data store and click on Sheet1 at the bottom left to proceed. This step creates a data extract which improves query performance.



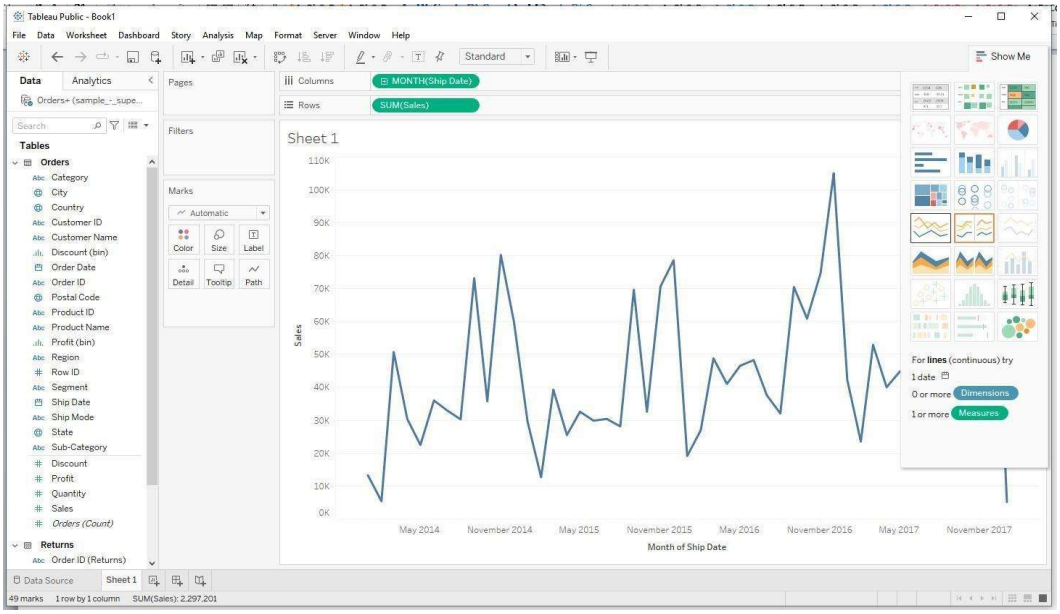
Sample Dataset:
https://help.tableau.com/samples/en-us/bookshop/Bookshop.xlsx

Data Visualization

We can perform various visualization operations on data in Tableau. Some of them are bar cart, histogram, bubble chart, gantt chart, scatter plot, heat map etc. Make variety of visualisations.

Line graph: Line graph can be continuous or discrete.

Continuous line graph is shown below:



Dashboards-

A dashboard is a way of displaying various types of visual data in one place. Usually, a dashboard is intended to convey different, but related information in an easy-to-digest form. And oftentimes, this includes things like key performance indicators (KPI)s or other important business metrics that stakeholders need to see and understand at a glance.

Dashboards are useful across different industries and verticals because they're highly customizable. They can include data of all sorts with varying date ranges to help you understand: what happened, why it happened, what may happen, and what action should be taken.

For example, category of sales across months in a year, region is the field added. The first view is shown below. This can be renamed at the bottom of the screen.

After creating individual views, now a Dashboard can be created by clicking on create dashboard at the toolbar.

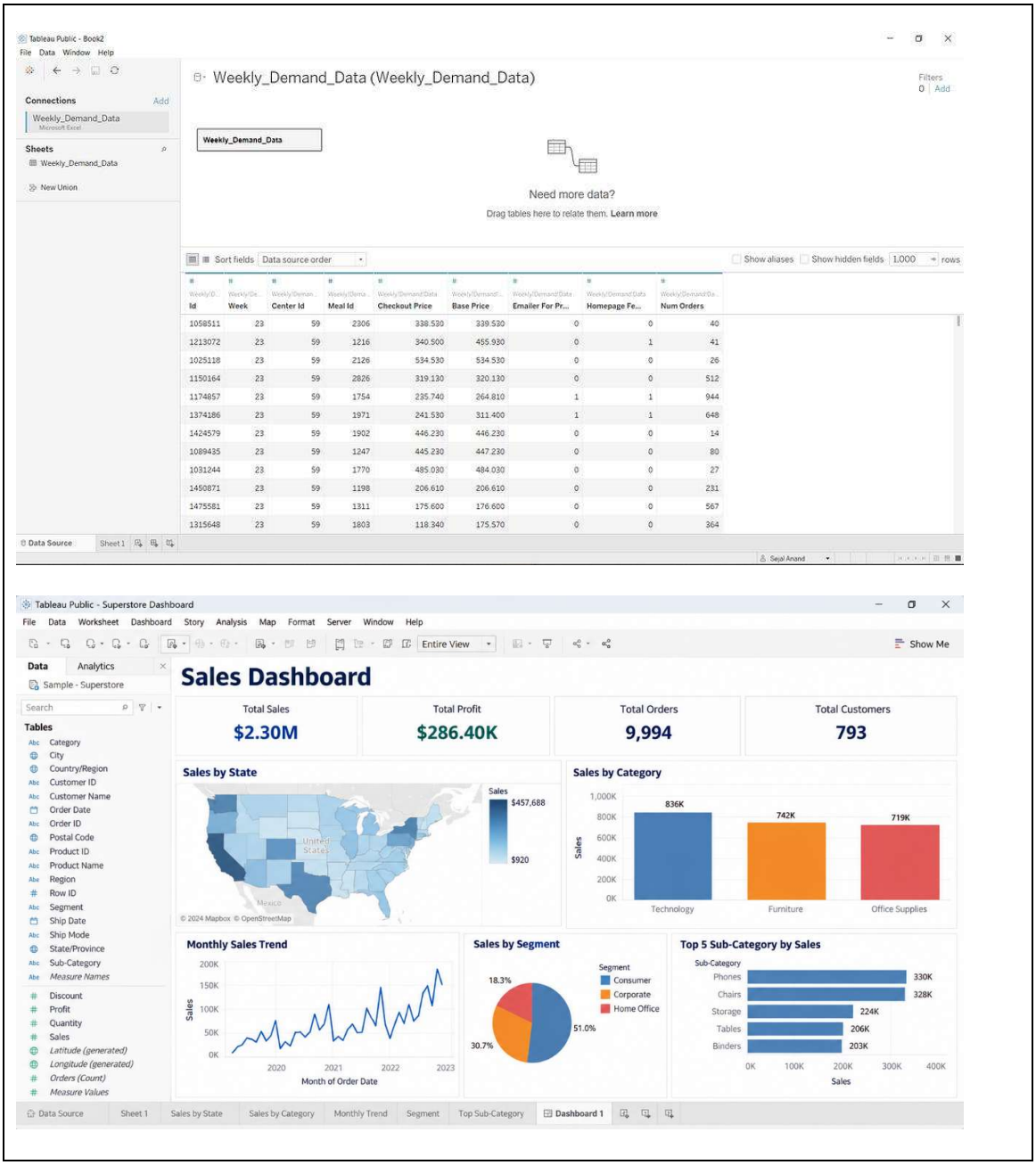
After clicking on new dashboard option, the screen is shown below.

now the sheets or views which are created earlier can be drag and dropped on this dashboard. The above three created views are placed in the dashboard as follows. One can follow their own way of importing sheets on the dashboard. After creating dahsboard, title can be given to the dashboard from Dashboard tab. Dahsboard can be customized in terms of its appearance by the user if requied. Dashboard once created can be saved on users system and can be retrieved whenever required.

Conclusion:

Thus, we successfully learned how to import different types of datasets, such as JSON and Excel/CSV files, into Tableau Public. We explored data preparation, relationships between tables, and various data visualization techniques. We also created a dashboard by combining multiple worksheets to present key insights in a clear and interactive manner.

Output /Screenshots:



Useful Resource:

https://www.tableau.com/data-model

https://catalog.data.gov/dataset/?res_format=JSON

https://www.tableau.com/resource/data-visualization

https://www.tableau.com/blog/relationships-tableau-data-model

https://help.tableau.com/current/pro/desktop/en-us/bookshop_data.htm

Exercise Questions

1. How does Tableau handle data from SQL databases?
2. How dataset is created in XML?
3. Create a scenario where a business analyst needs to import customer data from a SQL database and transaction records from a JSON API.
   Explain how Tableau can integrate both sources for unified visualization.

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① how does tableau handle data from SQL Database?
→ Tableau Connects to SQL databases such as MySQL, PostgreSQL, and Microsoft SQL Server using built-in Connectors. It allows user to import data or establish a live Connection to analyze data using Charts, Graphs, and dashboards.

② How is a dataset created in XML?
→ An XML dataset is created by Storing data in Structured format using user-defined tags. Each tag represents a data element, and element, and the data is organized in hierarchical Structure.

```
<?xml version="1.0" encoding="UTF-8"?>
<Customers>
    <Customer>
        <ID> 101 </ID>
        <Name> Rohit </Name>
        <city> Pune </city>
```

```
</Customer>
<Customer>
  <ID> 102 </ID>
  <Name> Amit </Name>
  <city> Mumbai </city>
</Customer>
</Customers>
```

③ Create a Scenario where a business analyst needs to import Customer data from a SQL database and transaction records from a JSON API.
Explain how tableau Can Integrate both Sources for unified visualization.

→ Scenario: A business analyst want to Combine Customer data from a SQL database with transaction records from a JSON API.

Steps:
ⓐ Connect Tableau to the SQL Database.
ⓑ Import transaction data from the JSON API.
ⓒ Join both datasets using Customer_ID.
ⓓ Create charts Showing Customer purchases and Sales trends.
ⓔ Combine the charts into an Interactive dashboard.